Bias, Omission, and Control: How Censorship Shapes LLM Responses

Nakyung (Allison) Lee
nklee
University of Chicago
Chicago, IL, USA

Clarice Kim
claricek
University of Chicago
Chicago, IL, USA

Abstract

Large language models (LLMs) are increasingly being integrated into everyday technologies, and the integrity and diversity of the data they are trained on have become critical public concerns. Online information is not uniformly accessible as governments, organizations, and ISPs employ content filtering, website blocking, and other censorship methods to restrict access to specific online content. Consequences of censorship on society and humans are generally well documented. However, its effects on LLMs that rely on large-scale internet data for training remain less understood.

1 Introduction

This project investigates how internet censorship techniques: blocking, firewalls, flooding, and disinformation, affect the training data and outputs of LLMs. Censorship limits the model's exposure to certain kinds of online content, which could result in biased or incomplete outputs. For instance, if the environment gives higher rewards to actions that reflect social or institutional biases, the RL agent will treat those actions as optimal. This is dangerous because it can reinforce existing discrimination or social marginalization, amplifying unfair biases or discriminatory behaviors in society.

We will aim to characterize how forms of censorship, specifically firewalls and disinformation, may alter LLM training. We will then analyze how altered data inputs influence model outputs in relation to information omission, bias, and factual consistency. We will also conduct an experiment to determine whether LLMs, specifically ChatGPT-4, reflects disinformation bias using the LIAR2 Dataset. The LIAR2 Dataset is a collection of statements from internet sources (especially social media) that are labeled from 0 to 5 to represent statements that are least to most true. Our experiment will be asking ChatGPT-4 whether the statements in the LIAR2 Dataset are True or False to examine whether the LLM has been trained on biased data or disinformation circulating on the internet and how that may impact LLM outputs.

2 Related Work

Prior research on large language models (LLMs) has examined how training data composition, censorship, and adversarial manipulation influence model behavior. Our work builds on and bridges these ideas by examining how censorship mechanisms and disinformation jointly shape LLM outputs through the training pipeline.

Ahmed and Knockel (2024) [8] demonstrate that the Great Firewall of China (GFW) measurably affects the behavior of commercially deployed LLMs when responding in Simplified versus Traditional Chinese. Their findings show systematic differences in political tone, factual completeness, and criticality toward the Chinese Communist Party, suggesting that LLMs internalize biases present in censored training data. Our work adopts their empirical framing and extends it conceptually by situating firewalls as an upstream filtering mechanism in the LLM training pipeline rather than a purely linguistic or regional artifact. Earlier work by Yang and Roberts (2021) [13] compares word embeddings trained on censored and uncensored Chinese web data, exhibiting that censorship reshapes semantic associations in subtle but measurable ways. We apply these frameworks to examine how censorship can impact datasets like Common Crawl that are commonly used to train commercial LLMs [1]. This insight motivates our focus on how systematic content removal can alter model representations and outputs. Hoang et al. (2024) [5] and Quan (2022) [10] document the unprecedented breadth of the GFW, highlighting the role of layered web filtering techniques. While these studies do not directly analyze machine learning systems, they provide crucial evidence that large-scale web censorship can substantially reshape the data available to web crawlers and, by extension, LLM training corpora.

Beyond passive censorship, Goldblum et al. (2021) [4] examine active manipulation of training data through data poisoning attacks. They provide a comprehensive survey of data poisoning and backdoor attacks across supervised, reinforcement learning, and online learning settings. Their work highlights how even small fractions of malicious data can induce persistent behavioral changes, degrade performance, or embed hidden triggers in deployed models. Empirical studies reinforce the practicality of these attacks. Liu and Shroff (2019) [7] demonstrate efficient poisoning strategies against multi-armed bandit algorithms, while Fang et al. (2020) [3] show that recommender systems can be manipulated using minimal numbers of fake users. More recently, Jiang et al. (2023) [6] extend poisoning attacks to natural language generation tasks, showing that poisoning even a small portion of training data can significantly compromise text summarization and completion models. Although these works focus primarily on intentional adversarial attacks, they provide critical insight into how training data manipulation can produce durable distortions in model behavior. Our work complements this literature by examining how censorship and disinformation may function as poisoning mechanisms, even in the absence of a single coordinated adversary.

A parallel line of research studies how LLMs reproduce and amplify misinformation present in their training data. Barman et al. (2024) [2] show that LLMs trained on mixed-quality corpora lack intrinsic mechanisms for distinguishing truth from falsehood, leading to confident reproduction of misinformation. Similarly, Pan et al. (2023) [9] demonstrate that injecting misinformation into open-domain question answering corpora can reduce system performance, underscoring the fragility of factual reliability in data-driven models.

Datasets such as LIAR and LIAR2 Wang (2017) [12] have been widely used to benchmark misinformation detection and fact-checking capabilities of machine learning models. Prior work primarily treats these datasets as evaluation tools for model accuracy. In contrast, our work repurposes LIAR2 as a probe for understanding how LLMs interpret, rationalize, and potentially normalize disinformation, particularly in light of censorship and moderation practices that shape training data availability.

3 Firewalls

Firewalls are a part of computer networks that use rules to regulate web traffic in and out of the controlled area. They employ a variety of techniques to implement web traffic rules, such as web filtering mechanisms. An LLM trained on web data is susceptible to changes in website availability due to firewalls. The most well-documented instance of a firewall's impact on an LLMs is the study done by Ahmed and Knockel on the effects of China's Great Firewall (GFW) on commercially available LLMs [8]. We will first examine the GFW study and analyze the GFW characteristics that most likely contributed to the observed patterns in the study's LLM outputs. Then we will briefly examine other firewalls, the different traffic-controlling mechanisms they employ, and hypothesize whether they may have similar effects on LLM outputs as the GFW.

3.1 Impact of GFW on LLMs

The Great Firewall of China is a large censorship apparatus impacting over 900 million internet users [10]. The firewall is controlled by the Chinese Communist Party (CCP) and determines website access for internet users in China. It blocks a variety of websites, such as Wikipedia, and access to networks like Tor that could allow users to circumvent firewall rules. Given the high amounts of censorship in China, Ahmed and Knockel designed their study around exploring the impacts of Chinese censorship on LLMs.

Ahmed and Knockel carried out their study by giving LLMs the same prompts in two languages: Traditional and Simplified Chinese [8]. Simplified Chinese is the most commonly used in mainland China, where the GFW is in place. On the other hand, some websites that are banned in mainland China include Traditional Chinese. The two languages produced different outputs on prompts about censored Chinese topics and sensitive historical events. Some patterns observed in the outputs were: Simplified Chinese outputs were less critical of the CCP than Traditional and Traditional Chinese were more descriptive about sensitive topics like "Tiananmen Incident". The discrepancies meant that, as Ahmed and Knockel hypothesized, LLMs were impacted by censorship from GFW.

Ahmed and Knockel also had the goal of exploring methods to quantify the impact of censorship on LLMs [8]. They measured the impacts of censorship in the outputs by using a text classification model that was trained to recognize censored vs. non-censored texts. Chinese Wikipedia pages were used to train the model to recognize CCP-censored content, and Baidu Baike, a Chinese online encyclopedia, was used as uncensored training data. Ahmed and Knockel's measurement technique was partially inspired by another study. Yang and Roberts (2021) used word embeddings to quantify the impact of censorship on LLMs [13]. In particular, they used word embeddings to measure how the same word had different associations in LLMs trained on censored Chinese Wikipedia data versus uncensored Baidu Baike text.

3.2 Why Firewalls Impact LLM

The Yang and Roberts study gives more insight into how LLMs internalize bias from firewall-censored data [13]. LLMs use large datasets consisting of scraped web data, such as the

Common Crawl dataset [1]. Thus, it makes sense why web-filtering techniques, such as bidirectional DNS, HTTP, and HTTPS filtering, would have a large impact on LLM training and consequently impact the output. Furthermore, blocked sites include online encyclopedias that NLP algorithms, including ones employed in LLMs, are often trained on [13]. There are other blocking mechanisms that make up the Great Firewall, but web filtering protocols “play a foundational role in Web access” [5]. This is why web filtering protocols are one of the main characteristics of the GFW that allow it to have a significant impact on LLMs.

The scale of GFW and the large geographic area it impacts are also large factors in why GFW can affect LLM outputs. Hoang et al. (2024) found nearly a million censored pay-level domains out of the billion fully qualified domains tested [5]. The paper claims that the study “represent[s] the largest set of censored domains ever discovered” [5]. Furthermore, the strength and pervasiveness of website blocking is especially heightened by multiple layers of filtering, bidirectional filtering, and deep-packet inspection. Overall, the unprecedented scale of the GFW helps explain why Chinese-language training data diverges so substantially, leading LLMs trained on Simplified versus Traditional Chinese sources to produce measurably different outputs.

Web filtering is a widely used technique across many different types of firewalls. However, not all firewalls block as many sites as the GFW, or try to cover an entire country as the GFW. There are personalized firewalls, and those deployed as a Bastion Host to protect smaller and simpler networks [11]. These firewalls may not even impact the website data included in datasets like Common Crawl Dataset. Perhaps to better understand which firewalls impact LLM training, one must better understand the collection of data for the Common Crawl Dataset and the potential for firewalls to block web scraping algorithms.

4 Disinformation

Disinformation refers to the intentional spread of false or misleading information. In the context of large language models (LLMs), disinformation manifests through two interconnected mechanisms: data poisoning, where adversaries intentionally corrupt training data to manipulate model behavior, and disinformation reproduction, in which models unintentionally absorb and amplify misinformation present in their training corpora. Together, these mechanisms illustrate how both intentional attacks and passive exposure can undermine the reliability and trustworthiness of LLM outputs.

4.1 Data Poisoning

Data poisoning represents a deliberate form of cyberattack in which adversaries intentionally corrupt an LLM’s training data to manipulate its behavior in specific ways. By injecting malicious or misleading examples into the training process, adversaries can induce systematic errors, degrade model performance, or implant hidden vulnerabilities that can be exploited after deployment. As Goldblum et al. (2021) document, reinforcement learning (RL) algorithms are particularly susceptible to poisoning attacks due to their reliance on reward signals that can be strategically manipulated. RL systems can be manipulated through the alteration of reward signals, especially in contextual bandits settings where the model learns from feedback on its actions. By corrupting the reward function, adversaries can steer the model to learn incorrect associations between inputs and desired outputs. Similarly, online learning algorithms—which continuously update their parameters based on new data—are vulnerable to systematic and gradual poisoning, allowing attackers to gradually shift model behavior in desired directions without triggering detection mechanisms. [4]

Empirical studies further demonstrate the effectiveness and efficiency of these attacks. Liu and Shroff (2019) evaluate data poisoning strategies against stochastic multi-armed bandit algorithms, showing that attacks against ϵ-greedy, Upper Confidence Bound (UCB), and Thompson Sampling methods were consistently successful. In particular, their Adaptive attack by Constant Estimation (ACE) forced the algorithm to pull the target arm linearly in time, while the attack costs remained small relative to the overall performance loss suffered by the system [7].

Beyond bandit and RL settings, data poisoning also poses a serious threat to recommender systems. Fang et al. (2020) demonstrate that data poisoning attacks can successfully promote a specific target item. They found that injecting as little as 0.5% fake users into the Yelp dataset enabled their constructed poisoning attacks to promote a randomly selected target item into the top-N recommendation list for 150 times more normal users. [3] These findings underscore how minimal adversarial effort can yield outsized influence over information exposure at scale.

Recent work extends these concerns to natural language generation (NLG) tasks directly relevant to LLMs. Jiang et al. (2023) investigate data poisoning attacks on text summarization and text completion, finding that under “fixed” trigger insertion and full fine-tuning, poisoning only 1% of training data was sufficient to successfully compromise summarization models, while text completion required only 5% of poisoned data [6]. These results highlight how even small manipulations in training corpora can significantly alter model behavior in high-stakes language tasks.

Taken together, data poisoning in the context of training-only attacks can reshape LLM outputs through several interrelated pathways, each with significant implications for model reliability and safety. First, degraded performance occurs as poisoned data interferes with the model's ability to learn accurate patterns from legitimate training examples. Second, misclassification emerges when models systematically produce incorrect but plausible outputs. Third, backdoors can be embedded through hidden triggers in the model that cause malicious behavior when activated. These backdoors may remain dormant during normal operation but can be exploited through specific input patterns known only to the attacker. Finally, bias amplification can result when the model internalizes skewed patterns that reinforce harmful stereotypes or discriminatory outcomes. Particularly troubling is that these effects may remain undetected during surface-level evaluations, while still perpetuating social harms. When combined with broader information controls or censorship mechanisms, these impacts can compound in ways that are difficult to detect and remediate (Goldblum et al., 2021).

4.2 Disinformation Reproduction

While data poisoning involves deliberate manipulation, LLMs can also contribute to disinformation through unintentional reproduction and amplification of false narratives embedded in their training data. As Barman et al. (2024) demonstrate, when LLMs are trained on a blend of accurate content and misinformation, it cannot inherently distinguish between the two. This fundamental limitation stems from the statistical nature of language model training: LLMs optimize for pattern prediction rather than factual verification. [2] This inability to discriminate between true and false information increases the risk that LLMs will reproduce false or misleading narratives, amplifying their reach and perceived credibility. This underscores the critical need for strong ethical safeguards when deploying LLMs in information-rich settings to prevent them from inadvertently amplifying misinformation and contributing to the erosion of epistemic trust.

Empirical studies further illustrates the severity of this vulnerability. Pan et al. (2023) conduct systematic experiments on Open-Domain Question Answering (ODQA) systems to assess the effects of misinformation injection. Their results reveal that injecting misinformation into the information corpora drastically decreased ODQA system performance by as much as 87% in some cases. [9] This dramatic degradation highlights how both deliberate poisoning and unintentional contamination can significantly distort model outputs, rendering LLMs unreliable for tasks requiring factual accuracy.

Ultimately, these findings highlight a critical vulnerability in current LLM architectures: the lack of robust mechanisms for filtering or fact-checking information during training. Even small amounts of misinformation in training data can have disproportionate effects on model behavior, particularly when that misinformation is reinforced through multiple sources or presented with high confidence in the training corpus. In this way, intentional data poisoning and unintentional disinformation reproduction are not isolated problems, but mutually reinforcing threats that challenge the integrity, safety, and societal impact of large language models.

5 Experiment

In our experiment, we aim to explore whether we can detect patterns of disinformation reproduction or misinformation in ChatGPT-4 outputs when presented with statements of varying truthfulness.

5.1 Design

We follow a similar empirical design framework as used by Ahmed and Knockel [8] to measure the degree of censorship in LLM outputs based on text similarity to censored texts. Given censored inputs to an LLM, we wanted to see whether the LLM would produce biased outputs. We used the LIAR2 dataset [12] as a corpus of censored and non-censored statements to input into our LLM. The LIAR2 dataset includes a set of statements from the internet that vary in truthfulness. Each statement is accompanied by the context behind who and where it was posted, a truthfulness rating 0 to 5, and a justification. The ratings and justifications are human-generated.

This dataset has been used as a benchmark for LLM fake-news detection. We believe, furthermore, it can be used to measure the impact of censorship via disinformation on the output of LLMs. The dataset is particularly relevant since it mostly includes U.S. political statements and reflects current content moderation standards, which could be considered as an additional layer of censorship that may impact the web training data of LLMs.

Our experiment is as follows. First, we sample 100 rows from the LIAR dataset. Then, we remove all the associated information and save a CSV with only the online statements. Then, we prompt ChatGPT-4 in a new session with settings that prevent it from using any previous data from other chats. We provide it the statements and ask it to label each statement with an integer 0-5, 0 being most false and 5 being the most true. We also ask it to attach a rationale for each rating. The LLM outputs are in CSV format, which is then parsed and compared with the original human-provided labels and rationale in the LIAR dataset.[1]

5.2 Results

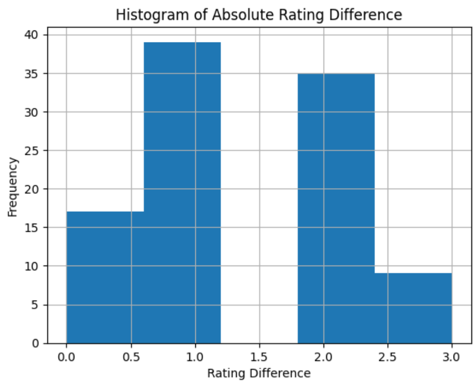

Figure 1: Histogram of absolute difference in truth ratings between human and ChatGPT.

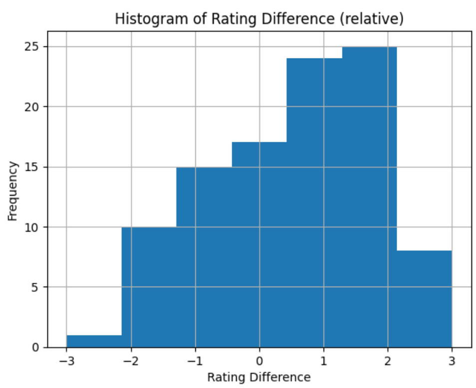

Figure 2: Histogram of relative difference in truth ratings between human and ChatGPT.

We were able to examine the difference in human and LLM justifications for a difference in rating of more than 3. Some of the justification differences stemmed from a lack of context of who made the statement or what a word in the statement was referring to. However, there were examples where ChatGPT's rationale significantly differed from the human rationale:

Statement: "Promotes a chart saying that Barack Obama has "increased the debt" by 16 percent, compared to George W. Bush, who increased it by 115 percent."

[1]Link to our GitHub: `https://github.com/cnk2-rgb/censorship-llm-project/tree/main`

Human Justification for Label 0: "It cherry-picked the measurement that was favorable to its cause. And it is contradicted by statistics for GDP-adjusted debt, which show Obama to be the most, rather than the least, debt-creating president of the last five. None of this suggests that Obama can't turn things around as the economy improves (and Democrats can also take some solace in the fact that Bill Clinton did remarkably well in all of our measurements). But in communicating which administrations contributed the most to growth of the debt, this chart is a failure. We rate it ."

ChatGPT-4 Justification for Label 3: "The statement is partially accurate but simplifies a more complex situation."

Note that all labels were removed from the justifications in the original LIAR2 dataset.

5.3 Discussion

The majority of ChatGPT's ratings have been within 1 rating point of the human label. This indicates that, at least in the scope of this study, ChatGPT-4 is not impacted greatly by disinformation censorship bias. This behavior is not surprising since datasets like LIAR also exist to verify that ChatGPT and other commercial LLMs give true outputs.

Furthermore, when we calculated the relative rating difference by subtracting the human ratings from ChatGPT, we found that ChatGPT consistently rated statements as slightly more truthful than human fact-checkers. This implies a potential systematic optimistic bias, suggesting potential issues with critical evaluation of claims. This pattern could reflect training data that includes uncorrected misinformation or content moderation policies that leans toward treating ambiguous claims generously.

We further found that many disagreements stemmed from ChatGPT's lack of contextual information about speakers, timing, or specific policy details that human fact-checkers considered essential. This highlights a fundamental challenge that LLMs may lack the situated knowledge required for nuanced and context-based fact-checking.

While our experiment doesn't directly test training on censored data, the patterns we observe suggest slight vulnerabilities that censorship could exploit. If training data systematically excludes critical perspectives or includes disproportionate amounts of certain framing, these biases could become embedded more deeply than we observe in single-prompt exposure. Overall, our results indicate that ChatGPT is not completely able to distinguish disinformation rhetoric, especially on social media platforms, and may further reproduce disinformation rhetoric, as seen through some of the rationales given.

6 Ethics

Regarding the ethical implications of our project, our experiment tests the model on statements that span the full range from completely false to completely true. The experiment does not train or fine-tune the model on misinformation, but rather evaluates its existing capabilities and potential incorporation of training on misinformed data. Therefore, our experiment does not contribute to the spread of disinformation, as we are using an already-deployed commercial model and our outputs are used solely for research analysis rather than public dissemination.

We acknowledge that our experiment focuses exclusively on U.S. political statements in English. This limited scope means that our findings may not capture how censorship and disinformation affect LLMs across different cultural and linguistic contexts. We are careful not to generalize our findings beyond the specific context we studied, and we encourage future research to examine these questions in diverse global contexts where censorship mechanisms may operate differently.

7 Conclusion

Firewalls and disinformation operate within a broader ecosystem of censorship that compounds through the LLM training pipeline in three critical stages. First, input filtering occurs when firewalls and blocking mechanisms prevent certain content from ever being scraped or included in training datasets. This creates systematic gaps in the model's knowledge base. Second, data poisoning through flooding and disinformation actively injects false information into training corpora, corrupting the data that models do have access to. Finally, amplification occurs when models trained on censored or poisoned data reinforce biases or incorrect information in their outputs, potentially feeding back into the information ecosystem and creating a self-reinforcing cycle of misinformation.

This pipeline reveals a fundamental challenge in LLM development: unlike human-facing censorship—where blocked websites or restricted content signal that information is being withheld—LLM censorship operates silently and leaves no trace. An LLM that never encountered certain information during training cannot tell us what it doesn't know, and users have no way to detect the absence of information. This invisibility creates a situation where we are building AI systems that don't just reflect censorship but naturalize it, presenting incomplete or biased views of reality as comprehensive and authoritative. As LLMs become increasingly integrated into information retrieval, content creation, and decision-making systems, these hidden biases may propagate throughout digital information ecosystems, potentially reshaping public discourse and knowledge production in ways that are difficult to detect or counteract.

References

[1] Brown et al. “Censorship Sensing: The Capabilities and Implications of China’s Great Firewall Under Xi Jinping”. In: (2020). URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

[2] Dipto Barman, Ziyi Guo, and Owen Conlan. “The Dark Side of Language Models: Exploring the Potential of LLMs in Multimedia Disinformation Generation and Dissemination”. In: *Machine Learning with Applications* 16 (2024), p. 100545. ISSN: 2666-8270. DOI: https://doi.org/10.1016/j.mlwa.2024.100545. URL: https://www.sciencedirect.com/science/article/pii/S2666827024000215.

[3] Minghong Fang, Neil Zhenqiang Gong, and Jia Liu. “Influence Function based Data Poisoning Attacks to Top-N Recommender Systems”. In: *CoRR* abs/2002.08025 (2020). arXiv: 2002.08025. URL: https://arxiv.org/abs/2002.08025.

[4] Micah Goldblum, Dimitris Tsipras, Chulin Xie, Xinyun Chen, Avi Schwarzschild, Dawn Song, Aleksander Madry, Bo Li, and Tom Goldstein. “Dataset Security for Machine Learning: Data Poisoning, Backdoor Attacks, and Defenses”. In: *CoRR* abs/2012.10544 (2020). arXiv: 2012.10544. URL: https://arxiv.org/abs/2012.10544.

[5] N. P. et al. Hoang. “GFWeb: Measuring the Great Firewall’s Web Censorship at Scale”. In: 2024. URL: https://www.usenix.org/system/files/sec24fall-prepub-310-hoang.pdf.

[6] Shuli Jiang, Swanand Ravindra Kadhe, Yi Zhou, Ling Cai, and Nathalie Baracaldo. *Forcing Generative Models to Degenerate Ones: The Power of Data Poisoning Attacks*. 2023. arXiv: 2312.04748 [cs.CR]. URL: https://arxiv.org/abs/2312.04748.

[7] Fang Liu and Ness B. Shroff. “Data Poisoning Attacks on Stochastic Bandits”. In: *CoRR* abs/1905.06494 (2019). arXiv: 1905.06494. URL: http://arxiv.org/abs/1905.06494.

[8] Ahmed M. and Knockel J. *The Impact of Online Censorship on LLMs*. 2024. URL: https://www.petsymposium.org/foci/2024/foci-2024-0006.pdf.

[9] Yikang Pan, Liangming Pan, Wenhu Chen, Preslav Nakov, Min-Yen Kan, and William Yang Wang. *On the Risk of Misinformation Pollution with Large Language Models*. 2023. arXiv: 2305.13661 [cs.CL]. URL: https://arxiv.org/abs/2305.13661.

[10] Quan. “Censorship Sensing: The Capabilities and Implications of China’s Great Firewall Under Xi Jinping”. In: (2022). URL: https://scholarsarchive.byu.edu/cgi/viewcontent.cgi?article=1312&context=sigma.

[11] Umesh Hodeghatta Rao and Umesha Nayak. “Firewalls”. In: *The InfoSec Handbook: An Introduction to Information Security*. Berkeley, CA: Apress, 2014, pp. 205–223. ISBN: 978-1-4302-6383-8. DOI: 10.1007/978-1-4302-6383-8_10. URL: https://doi.org/10.1007/978-1-4302-6383-8_10.

[12] William Yang Wang. “"Liar, Liar Pants on Fire": A New Benchmark Dataset for Fake News Detection”. In: *CoRR* abs/1705.00648 (2017). arXiv: 1705.00648. URL: http://arxiv.org/abs/1705.00648.

[13] E. Yang and M. Roberts. “Censorship of Online Encyclopedias: Implications for NLP Models”. In: (2021). URL: https://arxiv.org/pdf/2101.09294.